Cloud computing:

Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale. You typically pay only for cloud services you use, helping you lower your operating costs, run your infrastructure more efficiently, and scale as your business needs change.

Microsoft Azure Cloud:

Microsoft Azure is a comprehensive public cloud computing platform offered by Microsoft, providing 200+ services for building, deploying, and managing applications through a global network of data centers. It offers Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS) solutions, including computing, analytics, storage, and AI, primarily on a pay-as-you-go basis.

- Microsoft's cloud platform, similar to AWS and Google Cloud.
- Provides virtual machines, analytics, monitoring tools, and fast data processing.
- Eliminates the need for large physical infrastructure and investment.
- Cost-effective model where users pay only for the resources they consume.

Microsoft Azure Working:

Microsoft Azure is a public and private cloud platform that enables developers and IT professionals to build, deploy, and manage applications. It relies on virtualization, where a hypervisor abstracts hardware to run multiple virtual machines with different operating systems.

- Uses hypervisors to run multiple virtual machines on a single physical server.
- Supports operating systems like Windows and Linux within virtual environments.
- Operates massive data centers with interconnected servers to ensure scalability and performance.

Advantages of Cloud computing:

1. **Cost:**
    - Moving to the cloud helps companies optimize IT costs. This is because cloud computing eliminates the capital expense of buying hardware and software and setting up and running onsite datacenters—the racks of servers, the round-the-clock electricity for power and cooling, and the IT experts for managing the infrastructure. It adds up fast.
2. **Speed**
    - Most cloud computing services are provided self service and on demand, so even vast amounts of computing resources can be provisioned in minutes, typically with just a few mouse clicks, giving businesses a lot of flexibility and taking the pressure off capacity planning.
3. **Global scale**
    - The benefits of cloud computing services include the ability to scale elastically. In cloud speak, that means delivering the right amount of IT resources—for example, more or less computing power, storage, bandwidth—right when they're needed, and from the right geographic location.
4. **Productivity**
    - datacenters typically require a lot of "racking and stacking"—hardware setup, software patching, and other time-consuming IT management

chores. Cloud computing removes the need for many of these tasks, so IT teams can spend time on achieving more important business goals.

5. **Performance**
    - The biggest cloud computing services run on a worldwide network of secure datacenters, which are regularly upgraded to the latest generation of fast and efficient computing hardware. This offers several benefits over a single corporate datacenter, including reduced network latency for applications and greater economies of scale.
6. **Reliability**
    - Cloud computing makes data backup, disaster recovery, and business continuity easier and less expensive because data can be mirrored at multiple redundant sites on the cloud provider's network.
7. **Security**
    - Many cloud providers offer a broad set of policies, technologies, and controls that strengthen your security posture overall, helping protect your data, apps, and infrastructure from potential threats.

<u>5 Use cases:</u>

Here are five key use cases for cloud computing:

1. **Data Storage and Backup:**

   Organizations and individuals use cloud storage (like Amazon S3, Google Drive, or Dropbox) to securely store, sync, and back up large volumes of data across multiple devices, reducing reliance on physical, on-premise hardware.

2. **Testing and Development (Test/Dev):**

   Developers use cloud platforms to quickly provision, scale, and tear down testing environments. This allows them to build, deploy, and test applications faster without investing in physical servers.

3. **Big Data Analytics:**

   Cloud computing allows businesses to process and analyze vast, unstructured datasets to gain insights. Tools provided by cloud vendors (such as AWS, Google Cloud, or Azure) enable scalable analysis that would be impossible on a single machine.

4. **Software as a Service (SaaS):**

   Companies use web-based applications (like Salesforce, Microsoft Office 365, or Slack) that are hosted on the cloud. Users pay for access via subscriptions, removing the need for local installations and maintenance.

5. **Disaster Recovery (DR):**

   Businesses use cloud services for faster disaster recovery. Data can be backed up to multiple, geographically diverse cloud locations, enabling quick restoration of systems during a security breach or hardware outage.